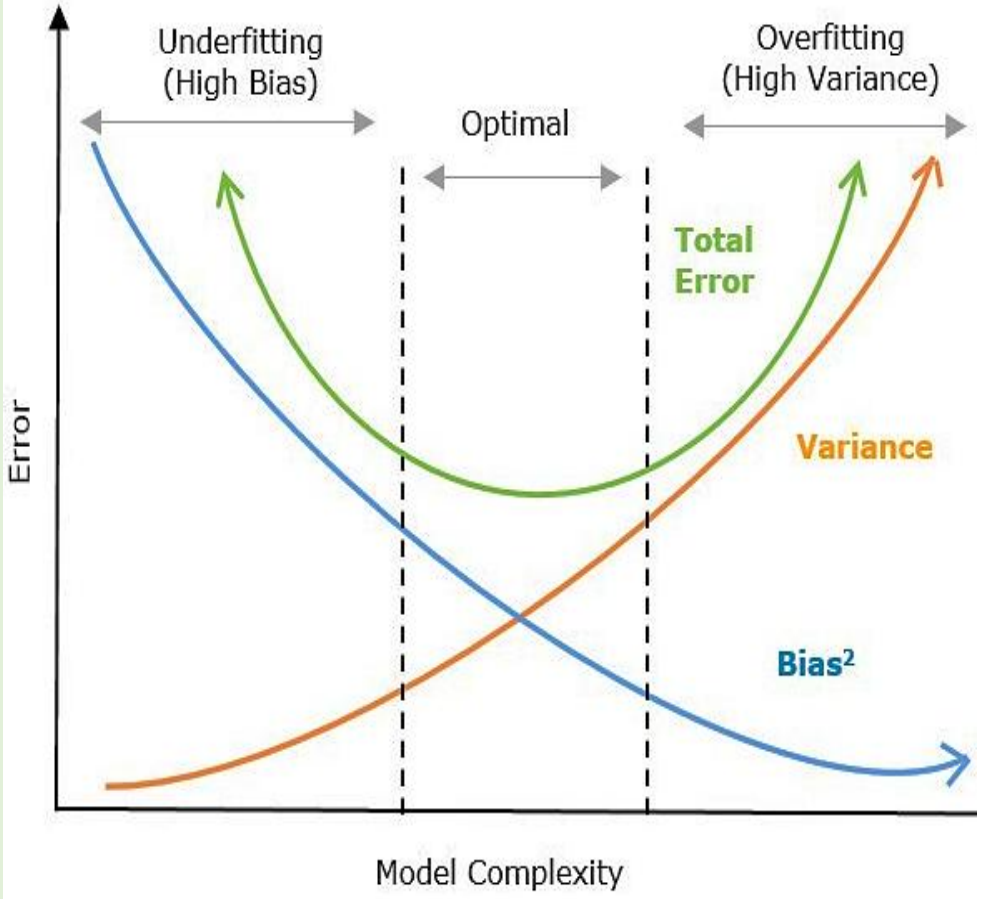
Deep Learning (23AID211)

Ensemble Learning

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Bias and Variance in ML/DL Models Training

- The bias-variance tradeoff is finding a balance between the error introduced by bias and the error introduced by variance.
- With increased model complexity, the bias will decrease, but the variance will increase.
- However, when we decrease the model complexity, the bias will increase, and the variance will decrease. So, we need a balance between bias and variance so that the total prediction error is minimized.
- When bias reduces, variance will increase. So we need to find an optimal bias and variance such that the prediction error is minimized.



What is Ensemble Learning

- Ensemble learning is a machine learning technique that aggregates two or more learners (e.g. regression models, neural networks) in order to produce better prediction accuracy, reduce variance, and enhance generalization to unseen data.
- It aims to mitigate errors or biases that may exist in individual models by leveraging the collective intelligence of the ensemble.
- This approach not only enhances accuracy but also provides resilience against uncertainties in the data.

Why use ensemble learning?

- **Reduce Overfitting:** By averaging multiple models, ensemble techniques can smooth out irregularities that may cause overfitting in a single model.
- **Improve Accuracy:** Combining multiple weak models leads to a more accurate prediction.
- **Increase Stability:** Ensembles are more robust to variations in the training data and less sensitive to noise.

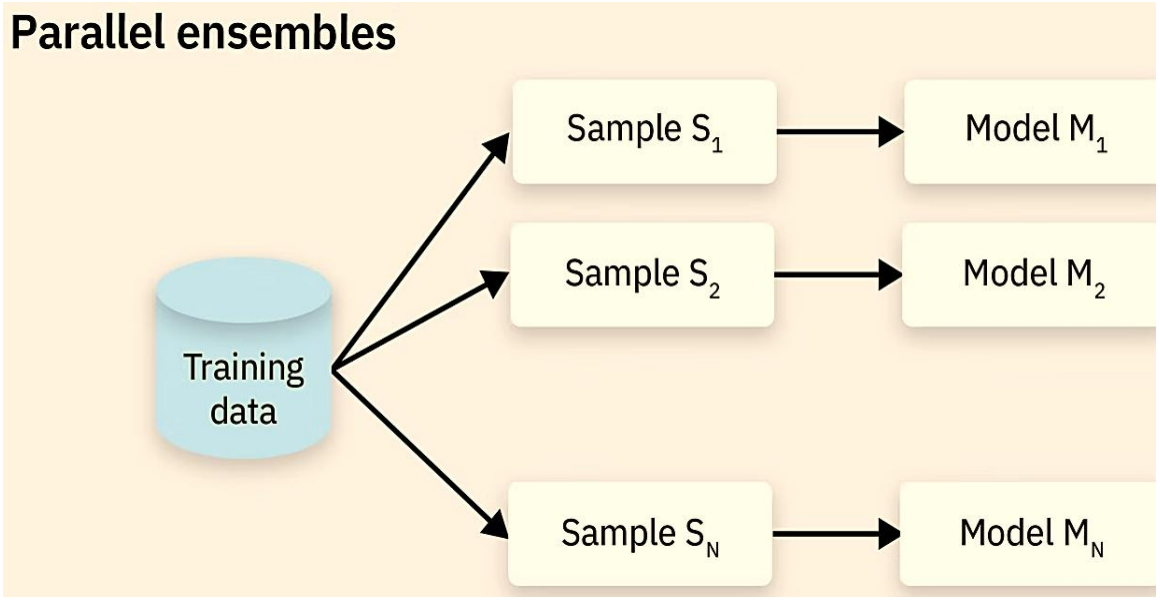
Types of ensemble models

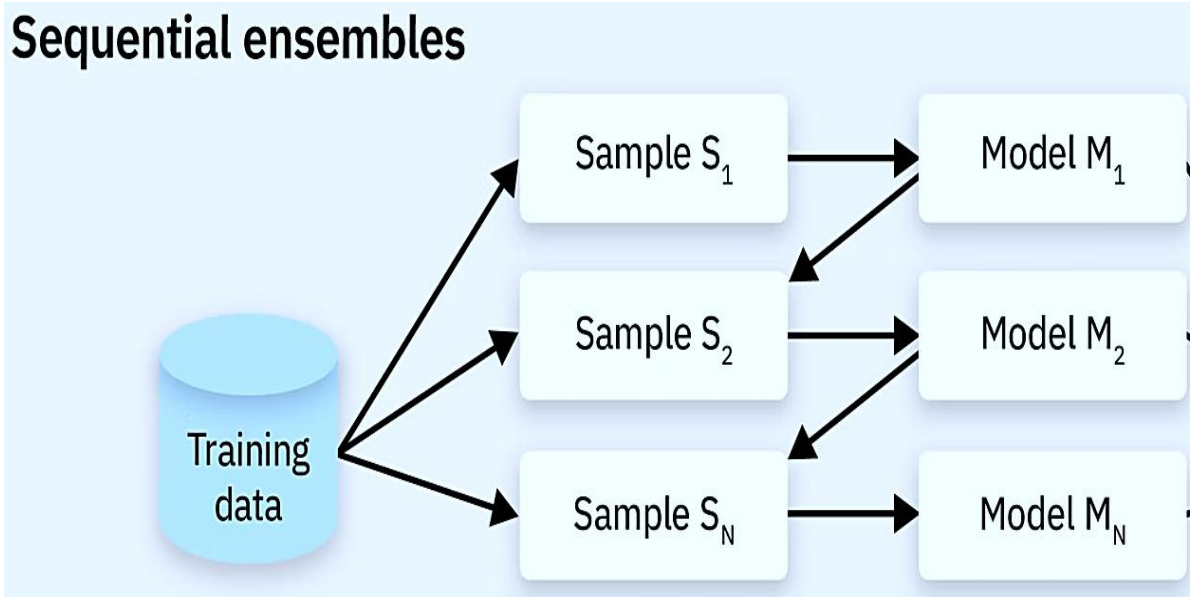
Literature widely categorizes ensemble learning methods in machine learning into two groups: parallel and sequential.

- **Parallel** methods train each base learner apart from the others of the others. Per its name, then, parallel ensembles train base learners in parallel and independently of one another.

- **Sequential** methods train a new base learner so that it minimizes errors made by the previous model trained in the preceding step. In other words, sequential methods construct base models sequentially in stages.

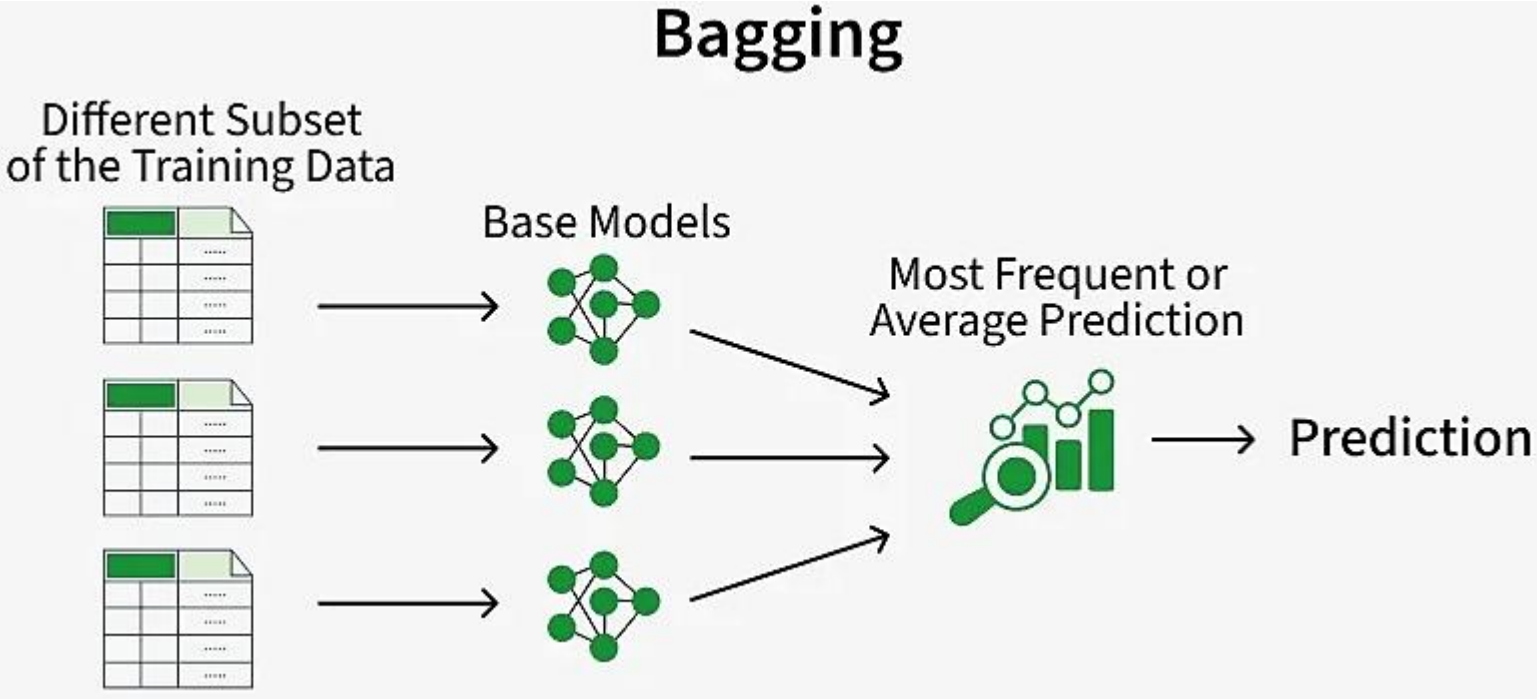
Types of ensemble models





Bagging (Bootstrap Aggregating)

- Bagging is an ensemble technique that aims to reduce variance and prevent overfitting by training multiple models independently and then averaging their predictions.



- Bagging works by generating multiple training datasets via bootstrapping and training a separate model on each.

Steps to Generate Training Datasets Using Bootstrapping:

1. Original Dataset: Begin with the original training dataset of size N, containing N data points or samples.

2. Sampling with Replacement: For each bootstrapped dataset, randomly select N samples from the original dataset. Since this is sampling with replacement, after selecting a sample, it is returned to the dataset, so it can be picked again in the future.

   - Some samples will be selected multiple times.
   - Other samples might not be selected at all in a particular bootstrapped dataset.

Steps to Generate Training Datasets Using Bootstrapping:

3. Create Multiple Bootstrapped Datasets: Repeat the above process multiple times to generate M different bootstrapped datasets, where M is the number of models (or weak learners) you plan to train in your bagging ensemble. Each of these datasets will have the same number of samples (N), but the distribution of data points within each dataset will differ due to random sampling.

4. Train Models: Train a separate model (e.g., a decision tree) on each of the bootstrapped datasets.

5. Aggregate Predictions: After all models are trained, combine their predictions (e.g., by majority voting in classification tasks or averaging in regression tasks) to make the final prediction.

Advantages of Bagging:

- **Reduces variance**: By averaging the predictions of multiple models, bagging reduces the model's variance, making it less likely to overfit.
- **Parallelizable**: Since each model is trained independently, bagging can be parallelised easily.

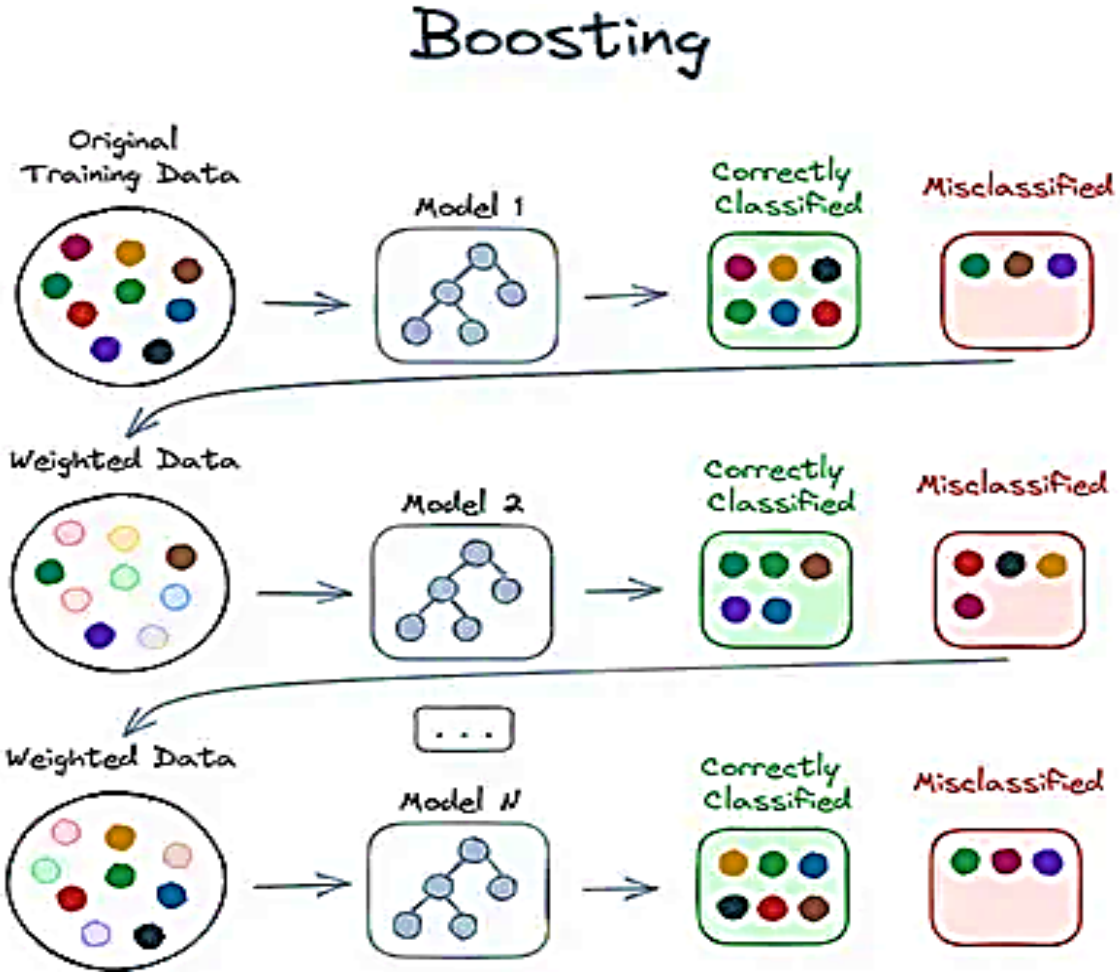
Disadvantages of Bagging:

- **Less effective on bias**: Bagging primarily reduces variance but does not significantly reduce bias if the individual models are weak.

Boosting

Boosting is another ensemble technique that focuses on **reducing both bias and variance** by training models sequentially, where each subsequent model attempts to correct the errors of the previous ones.

Unlike bagging, which trains models independently, boosting builds models iteratively.



How Boosting Works?

1. Train Weak Learners Sequentially: Models are trained sequentially, and each model pays more attention to data points misclassified by previous models.

2. Adjust Weights: In each iteration, the weights of incorrectly predicted samples are increased, so subsequent models focus more on those difficult-to-predict samples.

3. Aggregate Predictions: The final prediction is made by taking a weighted average or sum of the individual model predictions.

Advantages of Boosting:

- Reduces bias and variance: Boosting reduces both bias and variance, resulting in highly accurate predictions.
- Works well with weak learners: Even weak models, such as shallow decision trees, can be combined via boosting to create a powerful predictor.

Disadvantages of Boosting:

- Sensitive to outliers: Since boosting focuses on correcting errors, it can place too much emphasis on noisy data points or outliers.
- Computationally expensive: Boosting is a sequential process, making it slower than parallelizable methods like bagging.

Bagging meta-estimator:

The bagging meta-estimator serves as an assembling algorithm that can be used for both classification (Bagging Classifier) and regression (Bagging Regressor) problems. It follows the typical bagging technique to make predictions.

Steps for the bagging meta-estimator algorithm:

- Random subsets are created from the original dataset (Bootstrapping).
- The subset of the dataset includes all features.
- A user-specified base estimator is fitted on each of these smaller sets.
- Predictions from each model are combined to get the final result.

Random Forest:

Random Forest is another ensemble machine learning algorithm that follows the bagging technique. It is an extension of the bagging estimator algorithm. The base estimators in random forest are decision trees. Unlike the bagging meta-estimator, the random forest randomly selects a set of features to decide the best split at each node of the decision tree.

- Random subsets are created from the original dataset (bootstrapping).
- At each node in the decision tree, consider only a random set of features to decide the best split.
- A decision tree model is fitted on each of the subsets.
- Calculate the final prediction by averaging the predictions from all decision trees.

AdaBoost:

Adaptive boosting, or AdaBoost, is one of the simplest boosting algorithms. Usually, models use decision trees for modelling. The process creates multiple sequential models, with each model correcting errors from the previous one. AdaBoost assigns weights to incorrectly predicted observations, and the subsequent model aims to predict these values accurately.

- Initially, all observations in the dataset are given equal weights.
- A model is built on a subset of data.
- Using this model, predictions are made on the whole dataset.
- Errors are calculated by comparing the predictions and actual values.

AdaBoost:

- While creating the next model, assign higher weights to the data points that the model predicted incorrectly.
- Determine weights using the error value. For instance, a higher error assigns more weight to the observation.
- Repeat this process until the error function no longer changes or you reach the maximum limit of the number of estimators.

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Gradient Boosting:

Gradient Boosting, or GBM, is another ensemble machine learning algorithm that works for both regression and classification problems. GBM uses the boosting technique, combining multiple weak learners to form a strong learner. Regression trees are used as a base learner; each subsequent tree in the series is built on the errors calculated by the previous tree.



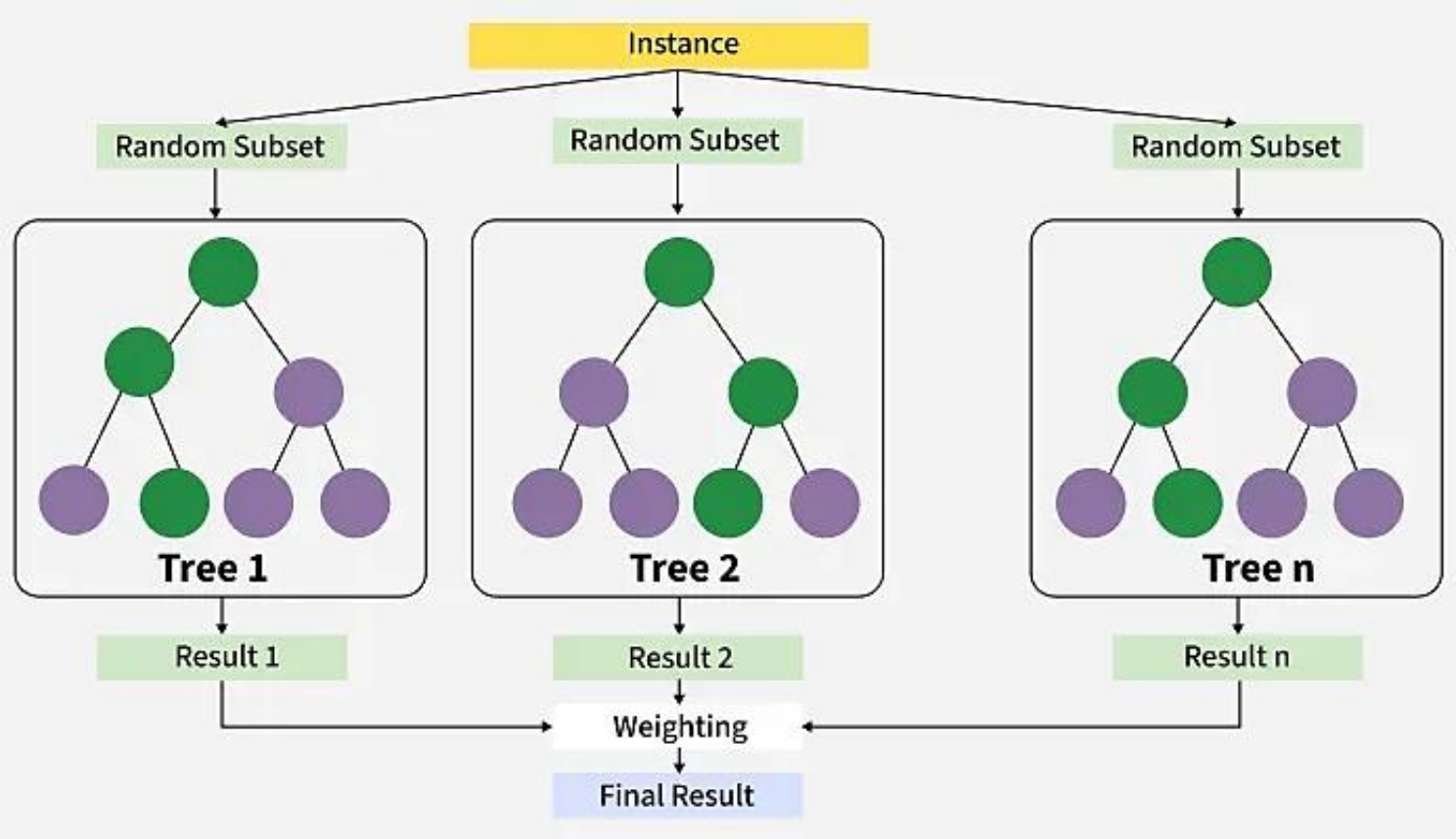
XGBoost:

XGBoost (extreme Gradient Boosting) is an advanced implementation of the gradient boosting algorithm. It has proved to be a highly effective ML algorithm, extensively used in machine learning competitions and hackathons. XGBoost has high predictive power and is almost 10 times faster than the other gradient boosting techniques. It also includes a variety of regularization techniques that reduce overfitting and improve overall performance. Hence, it is also known as the ‘regularised boosting‘ technique.

- Regularization
- Parallel Processing

XGBoost:

- High Flexibility
- Handling Missing Values
- Tree Pruning
- Built-in Cross-Validation



Evolution of Tree Algorithm

